

For presenters speaking to age 18 and high-school graduates



25

03

The artwork. The data. The choices between them.

Companion: [age-18-keyword-sheet.pdf](#)

Contents / Choose your route

Each script is complete. Background and stories follow the nine scripts.

Contents	2
Using this guide	3
M1 / Museum guide / 2-3 minutes	5
M2 / Museum guide / 5-10 minutes	6
M3 / Museum guide / 15-20 minutes	8
S1 / Science educator / 2-3 minutes	11
S2 / Science educator / 5-10 minutes	12
S3 / Science educator / 15-20 minutes	14
A1 / Artist talk / 2-3 minutes	17
A2 / Artist talk / 5-10 minutes	18
A3 / Artist talk / 15-20 minutes	20
Presenter background / The factual foundation	23
Source atlas / How much data, from where?	25
How the economic scale works	28
Methods / From text and points to the globe	29
Story collection / 25 ways to begin a conversation	33
Questions visitors may ask	40
Glossary / Words worth keeping close	41
References / Public sources and project evidence	42

Using this guide

Audience: age 18 and high-school graduates. Presenter edition. Last updated: 2026-09-09. Version: 1.0.

Nine spoken routes through the exhibition, with source notes and 25 country stories.

Before you begin

P.A.I.N. means Personal And Interconnected with Nature. The Personal-Planetary-Pain Map brings human and environmental indicators into one artwork. It invites attention and interpretation; its four layers do not form a validated universal measure of suffering.

Choose one voice and one duration. M is museum guide, S is science educator, and A is artist talk. Each script stands alone. Directions in shaded boxes are for you, not spoken words. Timings assume 120-140 spoken words per minute plus the explicitly timed pauses. Rehearse once: room acoustics, questions, and globe operation change delivery time. The longer scripts include complete explanations and observation periods; no separate story has to be inserted to finish them.

The scripts use a few rounded numbers. The background notes retain denominators, definitions, and references. Source IDs such as [L1] and [S1] are explained in the reference section. Public-source context is separated from what has actually been checked in the exhibition data.

The three routes

Route	Use it when	Where to begin
Museum guide	Visitors want an engaging tour of what they see.	M1, M2, or M3
Science educator	Visitors want to understand how evidence becomes an image.	S1, S2, or S3
Artist talk	Visitors want to explore the work's questions and metaphors.	A1, A2, or A3

All three routes explain the same selected exhibition: combined-v2 emotional data, 2024 GDP per capita, six physical conditions, temperature and emissions fields, and a Delaunay category network. Changing the voice changes emphasis, not the facts. Gabriel graphs are explained as the related alternative implemented during development. [L1-L5]

What every presenter should know

- The emotional input contains 249,438 analysed records from six sources and covers 192 countries.
- A displayed emotion is a model-selected category in sampled texts, not the identity of a people.
- The country profiles use the highest mapped physical, temperature, or emissions value inside each country polygon. Those values are not national averages.
- The socioeconomic layer is an artistic proxy based on GDP per person, not a poverty survey.
- Network lines express geographic construction within matching categories, not measured influence.
- Missing observations and a low or zero score mean different things.
- Original health and environmental processing details remain partly unverified. The notes say exactly which details. Do not turn those gaps into confident causal stories.



The numbered callouts identify: 1, emotional words; 2, the dotted and deformed physical surface; 3, atmospheric treatment; 4, the yellow socioeconomic signal; 5, layer controls. This is a capture of the running exhibition on 9 September 2026. Appearance depends on rotation, selection, theme, and ongoing rendering refinements. Colour and apparent size alone do not recover an underlying numerical value. The screenshots are illustrations of the artwork, not extra data.



Here the anger category is selected. 1 marks its category count, 2 the connecting arcs, and 3 the selected country label. The count of 152 refers to countries whose highest retained category is anger with all categories enabled. It is not a count of people or a percentage of angry residents. A later filtering choice can change category membership. [L1, L5]

M1 / Museum guide / 2-3 minutes

293 spoken words | estimated 2.1-2.4 minutes | planned pauses: 0 seconds

Start with all layers visible. Point to the globe, then the words, surface, atmosphere, and yellow areas.

Welcome to P.A.I.N. The name means Personal And Interconnected with Nature. This globe asks us to look at human suffering and the condition of our environment together. It combines art, data, and a way of drawing connections.

There are four layers. The words above countries belong to the emotional layer. A computer model has examined a selected collection of news, social posts, and complaints. It scores different kinds of distress, and the artwork displays the strongest retained category for each country. The word is a clue about those texts, not a description of everyone living there.

The physical layer draws on estimates for six painful conditions, including migraine and back pain. Population data helps place country-based information across the map. The surface turns that information into dents and dots. A dot does not stand for one patient.

The environmental layer brings together temperature change and emissions. Its haze is a visual metaphor: it is not a photograph of smoke. The yellow socioeconomic layer uses economic output per person. Lower values produce a stronger yellow signal, but money alone cannot describe a person's life or pain.

When we select an emotional category, lines connect countries sharing that category. The current network uses a geographic rule called Delaunay. The lines do not mean that one country caused another country's distress.

India is a useful place to look closely. Its physical signal is high, but the country symbol shows a peak among mapped points, not the average risk of an individual Indian. Population concentration and the way the data is placed both matter.

As you explore, ask three things: what am I seeing, what information produced it, and what is still missing? The artwork gives us a shared place to begin that conversation.

Invite one observation. Sources and numerical background: [L1-L5], [S1-S11].

M2 / Museum guide / 5-10 minutes

784 spoken words | estimated 5.9-6.9 minutes | planned pauses: 20 seconds

Begin on all layers. Pause briefly between layers so visitors can locate each visual cue.

Before we explain this globe, take a moment to notice what caught your eye. Perhaps it was a word, a bright country, a dent in the surface, or the haze around it. That first attraction is part of the work. P.A.I.N. brings several kinds of information into one image so that we can move between something visible and questions that are much harder to see.

The name means Personal And Interconnected with Nature. The artwork calls this globe the Personal-Planetary-Pain Map. It places bodily conditions, emotional language, environmental change, and economic circumstances together. Their coexistence is the starting point of the artistic conversation. It does not mean that all four have been measured with one common scientific instrument.

Let's begin with the physical surface. Its source attribution is the Global Burden of Disease study at IHME, combined with WorldPop population information. The exhibition contains six conditions: headache disorders, low back pain, migraine, neck pain, osteoarthritis, and rheumatoid arthritis. Each has 2,547 spatial records, making 15,282 records altogether.

Those records are not individual people. The project describes placing country-based health information using population data. Places where people are concentrated can therefore carry concentrated mapped values. The rendering then turns points into a continuous-looking surface, with overlapping dents and a dotted texture. The many visible dots are drawing elements, not a census of patients.

India gives us a practical example. Its physical country profile reaches about 0.943 on the map's normalized scale, with the peak coming from a migraine-labelled point. The country profile selects the highest point inside the boundary. It does not average all the people, all the cities, or all the conditions. That helps explain why the map can look intense around a densely populated place without establishing that every resident has unusually high risk.

Now look above the surface. The environmental layer has two separate inputs: temperature change attributed to Berkeley Earth, and emission-related values attributed to Climate TRACE. There are 8,100 temperature points and 2,339 emissions points in the exported map data. The red temperature treatment represents change, not simply how hot a place feels today. A cold region can experience substantial warming. Russia and Canada have high country-profile warming peaks in this dataset, which makes that distinction especially useful.

The emissions atmosphere is also an interpretation. It lets us see an otherwise abstract quantity, but the cloud is not measured smoke or local exposure. We still need the original processing details to identify the exact gas measure and period behind these exported values. We can explain what the source and the display are doing while being honest about that gap.

The yellow layer is more directly traceable. It uses World Bank GDP per person for 2024, expressed in current US dollars. GDP is economic production. Dividing by population gives a per-person comparison; it does not tell us what the typical person earns. The artwork compresses the enormous range using a logarithm, then reverses it so lower GDP per person gives stronger yellow. Burundi and Luxembourg make a useful contrast, but neither number contains a complete account of living conditions, inequality, or care.

Finally, consider the words. The selected emotional input contains just under a quarter of a million analysed records from six sources. News forms one group; five collections of social and complaint texts form another. The country score gives the news mean three quarters of the weight and the expression mean one quarter when both exist. Those are chosen weights, not the proportions of articles in the collection.

A custom classifier scores 14 distress categories. The highest retained category supplies the word above the country, translated through a country-language lexicon. Many categories can have some signal at once. A single visible word simplifies that much richer set of values. For news, the country usually identifies the publisher's origin, not where a reported event happened. We are looking at classified texts connected to countries, not a national opinion poll.

Select an emotional category. Allow 20 seconds to follow the network.

The connecting lines belong to a Delaunay network built from geographic positions within that category. Gabriel is a related, more selective rule that the project also explored. In both cases the connection is constructed from position and category membership. Its animation guides our attention; it does not show distress spreading through time.

The most useful habit is to hold two things together: curiosity about the image and care about what its data means. A blank country can mean missing information. A small sample can make an extreme score unstable. A country average, a local peak, and an individual experience are different things. With those distinctions in mind, the globe becomes a place to ask better questions about one another and the environments we share.

Optional discussion: invite a visitor to compare two layers over the same country. [L1-L5], [S1-S11]

M3 / Museum guide / 15-20 minutes

1,638 spoken words | estimated 15.2-17.1 minutes | planned pauses: 210 seconds

Allow the specified 210 seconds of looking and conversation. Keep those pauses even if the room is quiet.

Welcome. Before we put names and numbers to this work, let us give ourselves time to look. This is P.A.I.N., Personal And Interconnected with Nature. It asks how our personal experiences of suffering sit within the larger world that supports us. The globe is called the Personal-Planetary-Pain Map. It combines art, machine learning, geographic data, and a network that grows in response to our selection.

You may already recognize that this is an unusual map. Familiar countries sit on a surface that looks wounded. Words appear in different languages. A coloured atmosphere seems to cling to parts of the planet. The work gives those forms a shared space, while the information behind them comes from very different sources.

Looking pause: 30 seconds. Invite people to choose one feature they want explained.

We will follow four questions. What does each layer show? Where does its information come from? What choices turn information into an image? And what can a visitor reasonably conclude? These questions let us enjoy the artwork without assuming that its appearance is a direct measurement of everybody's suffering.

Start with the physical layer. The project attributes its health data to the Institute for Health Metrics and Evaluation's Global Burden of Disease study. That study produces health estimates from many kinds of evidence. Here we have six condition categories: headache disorders, migraine, low back pain, neck pain, osteoarthritis, and rheumatoid arthritis. Some names are familiar because we have experienced them ourselves; others may describe the experience of someone we know.

The map contains 15,282 spatial records for these conditions. There are 2,547 per condition. This tells us the size of the exported map input, not how many patients were consulted and not how many people have a condition. Those counts belong to different stages of the evidence. The source attribution says that country-based health information is distributed using WorldPop population data. This is a way to estimate where a national burden might be situated, rather than a separate health survey conducted at every dot.

The surface adds another transformation. Points create overlapping depressions; smoothing and rendering make them look like a continuous material. Dots help reveal that surface. They are visual samples generated by the artwork. A deeper-looking place can reflect both the input values and how nearby points accumulate. You cannot count the dots to count patients.

Let's use India to make this concrete. Its physical profile takes its highest mapped value from a migraine-labelled point, about 0.943 on a scale from zero to one. India has 266 contributing physical records inside the polygons used for this calculation. The country symbol shows the largest of those values, not their average. If we used an average or a population-adjusted rate, we would be asking a different question and might get a different ordering.

The population-based placement gives us a useful explanation for why crowded areas may appear concentrated. It does not, by itself, identify a medical cause. To explain an unusual migraine rate, for example, we would need to know the original health measure, its year, age structure, and uncertainty. Those details have not all been recovered for this export. We should keep the visible pattern and the medical explanation as separate steps.

Other countries make the distinction richer. Pakistan reaches a normalized peak of one in headache disorders and migraine. Japan reaches one in low back pain. China reaches one in osteoarthritis and rheumatoid arthritis. The condition behind a maximum matters. These values do not establish that the countries tie for an identical overall amount of suffering, because the original normalization across conditions has not been fully

reconstructed.

Looking pause: 40 seconds. Compare a strong physical region with a weaker one. Ask what extra information would make the comparison fair.

Move now from the surface to the atmosphere. The environmental layer has temperature-change and emissions inputs. The temperature source is attributed to Berkeley Earth. Its exported map contains 8,100 points. The emissions source is attributed to Climate TRACE, with 2,339 points. These numbers tell us the size of this map's input, not the size of either institution's complete database.

Temperature change is a surprisingly easy idea to misread. A place can remain cold while warming substantially compared with its earlier climate. A place that feels very hot today does not necessarily have the highest long-term change in this particular dataset. Russia and Canada have high country-profile warming peaks here. Research assessed by the IPCC describes amplified Arctic warming, with several interacting processes, including changes in snow and ice reflectivity. That is relevant regional context, not a reconstruction of every point on this map.

The country profile again uses a maximum. Russia is large and contains many contributing locations. More observed locations create more opportunities for an extreme value. The comparison therefore involves both a climate signal and a summary rule. A high local peak cannot simply be renamed a country's average warming.

The emissions atmosphere is equally important to read carefully. Climate TRACE publishes emissions estimates for different gases, sectors, and periods. The local category is called CO₂, but its exact original measure and time window still require confirmation. The visual haze is an artistic device. It is not a sensor reading of the air someone breathes, and its height does not represent a measured altitude. A dramatic cloud can start a conversation about emissions, but it cannot by itself tell us the exposure of nearby residents.

Looking pause: 30 seconds. Ask visitors to distinguish a change in temperature from an absolute temperature, using any familiar place.

The socioeconomic layer uses a quantity for which we can trace the displayed conversion: World Bank GDP per capita in 2024. GDP describes economic output. Per capita means divided by population. The units here are current US dollars per person. They are not purchasing-power adjusted and they are not the income of the median household.

The available values stretch from about 216 dollars per person for Burundi to about 288,002 for Monaco. A simple linear scale over that range would compress many countries near one end. The project uses a logarithm to make proportional differences easier to display, then inverts the result: lower GDP per person gives stronger yellow. Burundi sets the high endpoint of that artistic signal. Monaco sets the low endpoint. A zero at that endpoint does not mean that nobody in Monaco suffers.

Burundi and Luxembourg give us a useful story beyond the endpoints. The World Bank describes structural constraints on Burundi's economy, including low-productivity agriculture and landlocked geography. Luxembourg's national statistics office explains that cross-border workers contribute to production while the per-person denominator counts residents. These are reasons to understand the economic statistic before treating it as a description of individual experience. The artwork calls the signal socioeconomic pain, but the source variable itself remains economic output per person.

Now return to the emotional words. This layer is built through PCAI, which prepares and country-codes text, and pcai-classifier, which scores it. The selected exhibition uses 249,438 analysed records. Five thousand are from a news probe. The other 244,438 come from five expression collections: country-context tweets, civic-participation tweets, coordinate-linked texts, consumer complaints, and geotagged microblogs.

The dates and countries are unevenly covered. Most of the selected news records have recorded years 2015 or 2016. The full input is not a simultaneous global snapshot. News geography usually means the publisher's origin. A publisher in one country may report on events across the world. Other source collections assign

geography in other ways, such as profile locations or supplied country metadata. None of that is interchangeable with interviewing a representative group of residents about their current feelings.

The classifier learned from synthetic training examples. It then scores these real input texts for categories of distress. Different categories can overlap. A text can express both grief and anger. The country score averages those model outputs, with a chosen three-to-one weight between news and expressions where both are available. The artwork then selects the strongest of the 14 retained distress categories and displays a word through its language lexicon.

Here is why sample size belongs in a story. Palau has only one selected news record and no expression records. Burundi has 28 news records and two expressions. An extreme category score in either case demands attention to those small samples before a national explanation. The word above the country summarizes the chosen material; it does not speak for everybody.

Select anger and allow 40 seconds to follow the network from the highlighted origin.

The lines join countries assigned the same category. They use geographic label locations, not the locations of individual authors. The current rule is Delaunay: on the sphere the code constructs a network from the edges of a three-dimensional convex hull. Imagine a tight shell stretched around the category's points: the edges of that shell supply the Delaunay connections. The Gabriel rule, implemented during development, starts with those candidate edges and keeps a connection only when its diameter ball contains no other category point inside. We will use a small diagram later if you want to see that test.

The significant point for a visitor is simpler: this is a geometric construction. It does not measure communication, migration, friendship, shared policy, or the spread of emotion. The animation has a direction because someone selected a starting point. That direction is a way of revealing the image, not evidence of a historical sequence.

Try changing an emotional category filter. Removing a category does more than hide a word: the remaining scores can produce a different winner, so a country's network membership can change. The stored text scores have not changed. We have changed the question used to select what is visible.

Reflection pause: 70 seconds. Invite one observation, one possible explanation, and one thing the map alone cannot establish.

Take those three observations with you as you continue exploring the projection. That distinction is a good place to finish. The artwork makes distant conditions available to shared attention. The data gives us reasons to look, compare, and ask. Our responsibility as viewers is to notice the person who disappears inside an average, the place missing from a dataset, and the assumptions hidden inside a beautiful image. P.A.I.N. does not complete the conversation about care. It gives us a place to begin it together.

References: [L1-L5], [S1-S11], [C1-C4]. No country-causal claim is established by a map rank alone.

S1 / Science educator / 2-3 minutes

287 spoken words | estimated 2.0-2.4 minutes | planned pauses: 0 seconds

Point to one visible feature for each step: source, transformation, display.

Think of this globe as the end of several data pipelines. A pipeline begins with observations or estimates, applies a series of choices, and ends with a visual mark. To understand the mark, we need to follow that journey backwards.

The emotional layer begins with 249,438 selected text records from six sources. A trained model assigns scores to categories such as anger, grief, fear, and hardship. Country-level means are combined, then the highest retained category supplies the displayed word. The result describes model outputs on that collection. It is not the percentage of residents feeling that emotion.

The physical layer contains 15,282 map records for six health conditions. The project describes distributing country-based estimates using population information. The country profile selects the highest mapped value inside the boundary. A maximum is sensitive to where points are placed and how many opportunities there are to encounter an extreme.

Temperature change and emissions are separate environmental inputs. Their atmospheric appearance is a rendering choice, not a measurement of smoke thickness. The socioeconomic input is GDP per person in 2024. A reversed logarithmic transformation gives lower GDP values a stronger yellow signal. That choice does not turn economic output into a clinical pain scale.

The network adds another rule. Countries sharing the selected emotional category become nodes at geographic label positions. Delaunay geometry supplies the current edges. The related Gabriel rule keeps fewer candidate edges using an empty-diameter-ball test. Neither rule measures whether two countries influence one another.

There is a useful scientific question at every step: what is the unit, what is the denominator, and which choices could change the result? That is how we can appreciate this artwork while keeping its evidence and its interpretation clear.

Optional final question: what extra information would turn one of these comparisons into a testable claim? [L1-L5]

S2 / Science educator / 5-10 minutes

729 spoken words | estimated 5.7-6.6 minutes | planned pauses: 30 seconds

Keep the source-to-display diagram available. This script includes one 30-second reasoning pause.

We can read P.A.I.N. as a sequence of scientific and artistic decisions. Start with a source. Choose some of its records. Assign them to places. Calculate a summary. Map that summary to a visual feature. Every stage can change what the final image appears to say.

The emotional layer is a particularly clear example. PCAI prepares and country-codes source texts; pcai-classifier scores them. The selected exhibition combines 5,000 news records with 244,438 records from five social and complaint collections. This is a selected analytical product containing 249,438 records, not the entire upstream collection and not a count of unique people.

Geography requires a definition. For the news component, the country describes the publisher's organizational origin. It does not necessarily describe the article's subject. In the other collections, geography may come from a supplied country field, a location in profile metadata, or coordinate context. A country label therefore does not automatically mean the same thing across all six sources.

The classifier converts text into numerical representations and uses a trained prediction head to score distress categories. Training examples and analysed texts are separate: synthetic examples teach the model, while the selected real texts produce the exhibition's scores. The predictions remain diagnostic. They have not been validated as measurements of national emotional prevalence.

How do thousands of scores turn into one country value? First calculate a mean for news and a mean for the pooled expressions in that country. When both exist, multiply the news mean by 0.75 and the expression mean by 0.25, then add them. For an illustrative category with a news mean of 0.4 and an expression mean of 0.2, the result is 0.35. Those example numbers are invented to demonstrate the arithmetic. The weighting is a design choice, independent of how many rows each group contains.

The display picks the highest of 14 distress categories. Other outputs, including no-pain and out-of-scope scores, are excluded from this winner selection. That means a country receives a distress word even when an excluded control score is higher. A visible word cannot establish that most of the country's texts, much less most residents, express distress.

Reasoning pause: 30 seconds. If a country has just one sampled article, what would you want to know before explaining its score?

Now compare that pipeline with the physical layer. Its six condition surfaces use health estimates attributed to IHME and population-based placement attributed to WorldPop. We can inspect 2,547 map records for each condition. The original health measure and full allocation formula have not yet been recovered, so the exported intensity cannot responsibly be labelled as a prevalence percentage or a number of patients.

For a country profile, the code selects the maximum point inside its polygon, across the physical conditions. India has a peak near 0.943 from a migraine-labelled point. That is a statement about the current exported surface and the maximum rule. A national mean, a population rate, or an age-standardized comparison would be a different statistic.

Temperature and emissions also use point maxima for the country symbols. Temperature change is different from absolute temperature: a cold region can warm strongly. Emissions are different from the concentration of a gas in the air. The atmosphere around this globe is a way of drawing those inputs, not a measurement of what someone would inhale at that location.

The socioeconomic transformation is fully visible in the application. The input is World Bank GDP per person in current US dollars for 2024. The app takes its logarithm, rescales it between the available minimum and maximum, and reverses the result. Lower output per person therefore produces a larger displayed signal. A logarithm compresses ratios across a very broad range; it does not adjust for inequality or purchasing power.

Finally, the network uses countries sharing the displayed category. Delaunay edges are built from geographic positions on a sphere. Gabriel is a related option that tests whether another point lies inside the ball whose diameter is a candidate connection. The displayed animation reveals the network from a selected origin. No evidence of causal transmission enters that construction.

The scientific skill this artwork invites is asking what would change under another reasonable choice. Change a sample, a country rule, a mean into a maximum, or the set of allowed categories, and the image may change. Identifying those choices does not remove the artwork's meaning. It helps us locate that meaning accurately and ask which next observations would strengthen it.

Sources: [\[L1-L5\]](#), [\[S1-S11\]](#). *The arithmetic example is illustrative, not an observed country.*

S3 / Science educator / 15-20 minutes

1,648 spoken words | estimated 15.3-17.2 minutes | planned pauses: 210 seconds

Use the diagrams and the complete 210 seconds of discussion pauses. This is an explanation of the data pipeline, not a clinical interpretation.

Today we will investigate how this image is made. The useful starting point is not whether a country looks bright or dark. It is the chain of decisions between a source record and the mark we can see. P.A.I.N. combines several such chains, and understanding their differences is essential to interpreting the result.

Our working model has five stages: source, selection, geographic assignment, numerical summary, and visual encoding. We will follow the emotional layer through those stages, compare the other layers, and then examine the network. At the end, we should be able to name both a supported observation and a claim that would need more evidence.

The emotional source product contains 249,438 analysed rows from six collections. Five thousand rows come from a Stanford OVAL CCNews probe. The other 244,438 come from country-context tweets, DEMOTEC civic-participation tweets, Yachay coordinate-linked texts, CFPB complaint narratives, and CMU GeoText microblogs. Their sizes are highly unequal. The country-context tweet collection alone contributes 166,837 rows. CMU contributes 814. These are records, not unique participants.

The combined input has observations for 192 countries and 84 distinct language hints. A language hint is metadata, not proof of accurate classification in that language. The source years are also uneven. In the selected news probe, 4,692 of 5,000 records have recorded years 2015 or 2016. So a label on the globe must not be narrated as today's real-time national mood. Even the meaning of a date deserves care: a recorded year can depend on upstream metadata, and later source items need their publication-versus-crawl semantics checked.

Country assignment is the next decision. In the news component, country is based on publisher origin. Domain reference information and carefully handled country-code domains support that assignment. A news outlet in one country can publish a report about another. Consequently, the map's news country is not automatically the location of the suffering described.

Expressions use a range of existing geographic context. A supplied country field, a profile location, a coordinate bounding box, and a complaint location are different forms of evidence. They should not all be called an author's precise location. The pipeline selects eligible records under its geographic rules; the meaning of those rules must travel with the resulting country statistics.

Discussion pause: 40 seconds. Imagine a newspaper based in one country reporting on a disaster elsewhere. Which location would answer which question?

Now we reach the classifier. The system uses BGE-M3 text representations and a trained prediction head. It learns from synthetic examples, then produces saved predictions on the real selected input. Training and inference are different activities. A large training package does not mean that the same number of real people contributed to the exhibition.

The model first considers whether a text contains in-scope distress, then scores distress groups. In the public score composition, a group score combines the scope score with the conditional score for that group. For example, hypothetical scores of 0.8 and 0.5 yield 0.4 when multiplied. This is model arithmetic, not evidence that 40 percent of a population has a condition. The distinction matters because probability-like outputs can look more definitive than the validation actually supports.

The 14 groups can overlap. Anger, grief, and hardship are not mutually exclusive experiences. Their scores need not add to one. The model also has controls such as no-pain and out-of-scope, plus finer outputs not shown as the primary country word. Its saved acceptance status remains diagnostic: a completed computation is not the same as a validated national measurement.

The country aggregation has another important detail. It averages group scores across all selected rows in each cohort. It is not counting only rows above a classification threshold. For a country with both sources, the result is three quarters of the news mean plus one quarter of the pooled expression mean. Countries with only one cohort use that available cohort. This means that a small news sample can receive greater influence than a much larger expression sample. Weight represents a chosen importance, not sample proportion.

Let us work through invented numbers. Suppose a category's news scores average 0.4 and its expression scores average 0.2. The weighted country value is 0.75 times 0.4 plus 0.25 times 0.2: 0.3 plus 0.05, or 0.35. If we instead pooled all records equally, the result would depend on how many records each cohort contained. Those two methods answer differently weighted questions, even with exactly the same input texts.

Calculation pause: 40 seconds. Ask whether adding a thousand expression rows necessarily changes the chosen 75/25 weights. It does not.

There is a further distinction between numerical precision and evidential confidence. A saved score can have four decimal places because a computer calculated and rounded it that way. Those decimal places do not tell us how close it is to a human judgement. To assess that, we would need suitable independently labelled examples and a clear definition of the intended measurement. We would also need to consider whether the evaluation covers the languages, sources, and kinds of text used here. Repeating the same computation can establish that it is reproducible. It cannot by itself establish that the result represents national experience. This is why the guide retains the diagnostic status alongside its exact counts and formulas.

The displayed word adds a final selection. It takes the highest score among the 14 distress categories. It excludes the no-pain and out-of-scope controls from the competition. Palau shows why that matters. It has one selected news row, its displayed winner is uncertainty, and its out-of-scope score is higher than all the distress categories. The artistic word therefore cannot be treated as a finding that the country is predominantly distressed.

There is another useful intervention you can perform in the interface. Exclude a category. The stored scores stay the same, but the highest remaining category may change. The country's word and network membership can change with it. This is an excellent demonstration that a display depends on a question posed to the data, as well as on the data itself.

The physical layer follows a different chain. Its source attribution is IHME's Global Burden of Disease study, with WorldPop population information used to distribute country-based estimates. The six conditions contribute 15,282 spatial rows. The exact health release, measure, age and sex strata, population year, and normalization formula have not all been recovered. That prevents us from translating a normalized point directly into a prevalence rate, case count, or disability measure.

What we can verify is the country profile calculation: assign points to the displayed country polygons and keep the highest value. India has 266 physical records contributing to that calculation and a maximum of about 0.943. A point's contribution is not a patient count. If two countries have different numbers of mapped points, the country with more points has more opportunities to contain an extreme. This does not prove why either maximum is high, but it identifies a concrete comparison issue.

Comparison pause: 40 seconds. Contrast the largest exam mark in a class with its average mark. Then return to the country maximum.

The environmental layer has 8,100 temperature points and 2,339 emissions points. Temperature change needs a time window and reference climate; it is not the temperature on a particular afternoon. The project's source

panel describes a warming comparison and fixed reference values, but the original processing needs confirmation before we reverse normalized scores into degrees. A scientific explanation should identify that limit instead of manufacturing precision.

Emission quantity also requires a gas, a period, and an accounting scope. Climate TRACE provides several such choices. The local category name CO2 is not enough to establish which one produced the export. Moreover, emissions, atmospheric concentration, and personal exposure are distinct quantities. The hovering form is a visual encoding, not another measured field of air quality.

The socioeconomic source is easier to follow exactly. It is World Bank GDP per capita in current US dollars for 2024. There are 190 available values in the selected roster. Missing entries are omitted from numerical rendering rather than treated as zero. The application maps the logarithm of each value between the observed minimum and maximum, then subtracts that fraction from one. The chosen endpoints are Burundi and Monaco.

Taking a logarithm compresses multiplicative differences. Inverting the scale makes low GDP per person look strong in the artistic pain layer. Neither operation adds information about income distribution, household costs, social support, or subjective suffering. Combining this layer visually with health and climate does not validate a formula that adds their scores. The same numerical range does not give different measures the same meaning.

Let us finish the construction with the graph. Each node is a country label location converted to a point on a unit sphere. The current winning emotional category assigns the node to a group. Within each group, the Delaunay construction uses the edges of a three-dimensional convex hull. The result is rendered as curved great-circle arcs lifted above the globe. Imagine a tight shell stretched around the category's points. Its edges give us the connections before the renderer curves them above the surface.

Gabriel is a related alternative. For each candidate edge, form a ball whose diameter joins its endpoints. If another point in that category lies strictly inside the ball, reject the edge. This produces a more selective proximity network. The application has a small-group special case described in the notes. Neither construction uses correlations between time series or evidence of interaction between the countries.

Diagram pause: 40 seconds. Follow the accepted and rejected edges in the Gabriel illustration. Then distinguish construction from what a line might make us imagine.

The animation reveals edges outward from the selected origin using graph distance. That sequence makes the network easier to follow, but it does not introduce a time model of emotional contagion. Similarly, font size, label brightness, and atmospheric thickness contain readability and rendering choices. They should not be mistaken for additional calibrated measurements.

Closing pause: 50 seconds. Invite one supported observation and one additional dataset or test that would be needed to evaluate a causal explanation.

We now have a way to read the globe with both interest and precision. Every visible pattern has a source, a selection rule, a geographic interpretation, a summary, and an encoding. We can question each stage without dismissing the artwork. Scientific curiosity begins with those distinctions: knowing what has been measured, what has been constructed, and what we would need to learn next.

References: [L1-L5], [S1-S11]. Both arithmetic examples are hypothetical teaching examples.

A1 / Artist talk / 2-3 minutes

293 spoken words | estimated 2.1-2.4 minutes | planned pauses: 0 seconds

Let the globe remain visible while you speak. Give the final invitation a short silence.

A map usually invites us to name places. This one invites us to notice wounds, words, and relationships. P.A.I.N. means Personal And Interconnected with Nature. It asks us to imagine our lives within the living systems around us, while keeping sight of how differently pain is experienced.

The globe has four layers. Physical health estimates are translated into a marked and dotted surface. Emotional language from selected texts appears as words above countries. Temperature change and emissions are given an atmosphere. Economic output per person becomes a yellow signal, stronger where that output is lower.

These translations are part of the artwork. The planet's pain is a metaphor through which we can consider harm and care. It is not a claim that the Earth has a human nervous system, or that four different datasets can measure suffering in the same units.

The words deserve particular care. A model has read a selected collection of news, posts, and complaints and scored categories of distress. The chosen word above a country is not the voice of all its residents. It is an invitation to ask whose language has entered the image, whose experience may be absent, and what happens when many voices become one label.

When a category is selected, lines appear between countries sharing it. The current network uses a geographic Delaunay rule. Its connections are constructed, not discovered pathways along which pain has travelled. Yet seeing a familiar word repeated in another language can still make distance feel different.

You do not have to offer a personal story to participate. You can begin simply by looking. Choose one place, notice what the artwork brings forward, and consider what a fuller account of care for that place would need to include.

Presenter note: this is an original exhibition narration, not a quotation attributed to an artist. [L1-L5]

A2 / Artist talk / 5-10 minutes

737 spoken words | estimated 6.3-7.1 minutes | planned pauses: 60 seconds

Start with all layers. Include a 30-second silence after the first invitation and a 30-second pause near the end.

Take a moment to find a place on this globe. It might be somewhere you live, somewhere you have never visited, or somewhere associated with a person you care about. Notice what the image offers you: a word, a surface, a colour, a surrounding atmosphere. Notice also how little of that place can fit inside any of those forms.

Looking pause: 30 seconds. No personal disclosure is requested.

P.A.I.N., Personal And Interconnected with Nature, brings human and planetary conditions into one space. The four layers ask us to move between bodies, emotional language, environmental change, and economic circumstances. The artwork is interested in their relation. Its data does not provide a single instrument for weighing one person's suffering against another's.

The surface is a translation of health information. The source attribution names IHME's Global Burden of Disease study and WorldPop population information. Six painful conditions enter the physical layer, including migraine, back pain, and arthritis. Country-based information is placed through a population-based process, then rendered as a surface that seems to carry traces of strain.

Calling those forms scars is an artistic choice. The texture gives something abstract a material presence. It does not show the actual shape of anyone's wound. Its many dots are generated for the drawing; they are not individual people. This distinction lets the image remain suggestive without claiming an intimacy the data does not possess.

India is a place where that suggestion can be powerful. A high mapped physical value catches our attention, but the country profile represents the largest point in its boundary. It is not the voice of a billion individual experiences, and it is not a national average. The population-based placement gives us one reason to think about concentration. The medical reasons behind an individual's condition require a different kind of knowledge.

The atmosphere translates another set of relationships. Temperature change becomes a red treatment; emissions become another atmospheric form. These are invitations to perceive quantities that are hard to hold in mind. A cloud in the artwork is not a photographed plume. Its height and density are not a direct measurement of what someone breathes.

A cold region can carry a strong warming signal. That reversal is worth pausing over: what we feel on a particular day and what a long-term comparison reveals are different ways of knowing a place. The map brings a historical change into the present moment of looking, without turning every visible mark into a complete environmental explanation.

The yellow layer uses GDP per person. Lower economic output per person produces a stronger artistic signal through a reversed logarithmic scale. It is a deliberately limited proxy. It cannot tell us who has access to a home, who receives care, who owns wealth, or whose work has been left uncounted. Those questions are not answered by the colour. They are questions the colour can help us remember to ask.

The emotional words carry a different kind of tension. Their input includes news, social posts, and complaints. A model scores forms of distress in that selected material. The exhibition displays a winning category in a chosen language for each country. There are 249,438 analysed records and 192 countries with observations, but the material is uneven across places and years. It is not a chorus in which every person has been given equal time.

Think about the difference between an experience and a word used to describe it. Think about the further difference between that word and a model score. Each step makes comparison possible, while leaving something behind. Translation can bring a word closer to one visitor and make it less familiar to another. One chosen country language also cannot represent every language spoken there.

Select a category. Allow 30 seconds to follow its lines without narration.

The current network is built by a geographic Delaunay rule, joining countries that share the displayed category. The project also explored a more selective geometric rule called Gabriel. These are constructed connections. Their animation does not reveal where an emotion began or how it moved. The lines nevertheless offer a way to look across distance, guided by a shared category rather than by the borders alone.

We can leave with a question rather than a verdict about any country. What does the image make available to attention? What is still missing? What form of listening, research, or care would help us understand the people and environments behind it? That unfinished relationship is part of the work, and we are invited to carry it beyond the projection.

Original narration. Sources and display boundaries: [L1-L5], [S1-S11].

A3 / Artist talk / 15-20 minutes

1,627 spoken words | estimated 15.6-17.6 minutes | planned pauses: 240 seconds

Include the full 240 seconds of observation and response. The invitation is reflective, never a request for visitors to disclose personal pain.

Let us begin with the experience of looking at a globe. It makes the world seem available to us all at once. We can see a boundary, travel across an ocean with our eyes, and name a country without knowing the life of a single person there. That power is useful, and it is also a limitation. A world that fits inside an image has already been simplified.

P.A.I.N. begins inside that tension. Its name means Personal And Interconnected with Nature. The Personal-Planetary-Pain Map brings four kinds of information together: physical health, emotional language, environmental change, and economic circumstances. It uses their meeting as an invitation to think about bodies and the conditions that sustain them.

Before we examine the sources, choose a place to look at. It may be familiar or unfamiliar. Notice whether a word reaches you first, or a colour, or a shape. You do not have to tell us why you chose it. Observation is enough to participate.

Looking pause: 40 seconds. Allow a visitor to point without explaining a personal connection.

The first layer is physical. The artwork turns health information into a marked surface. It gives statistical estimates a texture that can look strained, wounded, or scarred. Those are metaphors in the material language of the artwork. They are not photographs of bodies, and they do not identify the location of an individual person's wound.

The source attribution names the Global Burden of Disease study at IHME, together with WorldPop population information. The six conditions are headache disorders, migraine, low back pain, neck pain, osteoarthritis, and rheumatoid arthritis. The map uses 15,282 spatial records, with 2,547 for each condition. This count describes the exported data that enters the visualization, not a count of people who consented to tell a story here.

The project describes distributing country-based health information according to population. That is one way to give a national estimate a location. It brings the possible concentration of a burden into view, while also making an assumption about how information should be placed. The drawing then adds overlapping depressions, smoothing, and a dotted surface. The dots help us see the form. They should not be mistaken for a series of individual lives.

When we look at India, this distinction becomes especially important. Its high physical profile value comes from a peak in the mapped migraine surface. The symbol keeps the largest value inside the country's polygon. It does not summarize every person by an average, and it cannot tell us how one person's pain compares with another's. A population-based placement can help explain concentration in the image. The clinical reasons for illness need evidence beyond that placement.

We might be tempted to turn a dramatic mark into a dramatic national story. The more careful invitation is to ask what the mark has made visible and which lives remain unavailable to this form of representation. A faint area does not tell us that life there is easy. It may reflect the source, the coverage, the normalization, and the scale at which we are looking.

Reflection pause: 40 seconds. Ask what a texture can communicate that a table cannot, and what a table can preserve that a texture may hide.

The environmental layer asks us to look at the space around the surface. One input concerns temperature change and is attributed to Berkeley Earth. Another concerns emissions and is attributed to Climate TRACE. The artwork gives them an atmospheric presence. We are looking at a visual translation of quantities, not a photograph of the air.

Temperature change has a particular relationship to memory. It compares a place with an earlier condition rather than telling us only how it feels today. A cold region can experience substantial warming. Russia and Canada have high warming peaks in the map's country profiles. The point of that example is not to name a winner in a ranking. It is to notice that a familiar image of a cold place may conceal an unfamiliar history of change.

The atmosphere's shape is part of the artwork's expression. Its height is not a measured altitude. Its thickness is not a direct measure of exposure. The emissions export is labelled CO₂, but its exact original gas measure and time period still require confirmation. That uncertainty does not need to disappear for us to look meaningfully. It needs to remain visible in the explanation, so the image does not speak with more certainty than its source.

The phrase planetary pain is similarly an invitation to interpretation. We are not claiming that the Earth has a nervous system like a person's. The metaphor lets us consider harm, dependence, and care across scales. A forest, a body, an economy, and a community can all be affected by change, but their experiences and mechanisms are not interchangeable.

There is something productive about bringing those differences close without erasing them. An artwork can hold together questions that are separated in a database. We can explore relations without pretending that every relation has already been measured, or that every line in an image establishes a causal link in the world.

Looking pause: 40 seconds. Let visitors compare surface and atmosphere without deciding that one explains the other.

The socioeconomic layer adds a yellow signal. Its selected source is World Bank GDP per person for 2024. GDP measures economic output. A per-person figure divides that output by population. The artwork uses a logarithmic transformation, then reverses the direction so that lower output per person appears stronger in the pain layer.

This translation is deliberately open to questions. A measure of production is not a measure of affection, security, fair distribution, or dignity. The same country figure can sit above people with very different lives. The map cannot tell us the experience of the person whose income differs most from an average, whose work is poorly recognized, or whose suffering does not follow the economic comparison.

Burundi and Luxembourg can anchor that discussion. The selected GDP-per-person values are about 216 dollars and 137,782 dollars respectively. Those figures are important economic information, but they do not provide two complete portraits of human well-being. The point of putting the countries beside one another is to ask what an economic indicator illuminates and what else an account of care would have to include.

Monaco supplies the high-GDP endpoint used by the artwork's reversed scale. Its resulting yellow signal is zero. That is a statement about a chosen mathematical reference, not an announcement that there is no suffering in Monaco. A missing value elsewhere is different again: it means that this selected file does not supply the observation. Absence of data should not be filled with a comforting story.

Now consider the words that seem to hover between the place and the viewer. The emotional layer draws from a selected collection of news, social posts, and complaints. PCAI prepares and country-codes those texts. A custom classifier scores categories of distress. The artwork then chooses the strongest retained category and presents it through a country-language lexicon. The visible word is the last step of a long translation.

The selected product contains 249,438 analysed records from six sources. These records are uneven across countries, languages, and years. Much of the small news probe has recorded years 2015 or 2016. For news, the country generally names the publisher's origin; it may not name the place described in the article. The image

therefore offers a view of classified material connected to places, rather than the current emotional voice of every population.

One word cannot contain the many forms of distress that may occur together. Grief and anger can coexist. A word translated as hardship may carry associations that differ between languages. The chosen language for a country may be familiar to some residents and not to others. These are reasons to keep translation open to conversation, rather than treating the lexicon as the final meaning of a culture.

There is a small country example that makes the limits concrete. Palau's emotional result comes from one selected news record. Its displayed word is uncertainty. Before telling a story about uncertainty in Palau, we should recognize how small the sample is and how much the selection rule contributes. The artwork has made a word visible. It has not given us permission to speak on behalf of a country.

Word pause: 40 seconds. Invite visitors to consider a word in a language they know. Ask what a brief translation might leave out.

When we select a category, the work draws connections. A country becomes a node at its geographic label position. Countries with the same selected category are grouped, and the current Delaunay rule supplies edges between positions. Curved lines lift that structure off the surface so that we can follow it with our eyes.

The project also explored Gabriel graphs. Their more selective geometric test keeps a candidate connection when the ball spanning its endpoints contains no other point of that category inside. That mathematical choice changes the pattern of lines. It does not add evidence that two populations share an experience in the same way.

It is possible for a constructed line to move us without mistaking it for a discovered relationship. A word we recognize may appear elsewhere in another script. A selection may bring a distant place into attention. We can value that experience and still say plainly that the network does not measure communication, contagion, migration, or influence.

The animation begins from a selected origin because the viewer needs somewhere to start. Its sequence is not a chronology of suffering. If we change the category filter, a country's winning word can change even though its underlying scores do not. The image responds to the question we ask of it. That responsiveness is part of its artistic form.

Closing pause: 80 seconds. Invite an observation about connection, an unanswered question, or a thought about care. Passing is welcome.

We can finish where we began, with the difference between seeing a world and knowing a life. P.A.I.N. makes information available to attention through texture, language, atmosphere, and connection. It also leaves us with the responsibility to look beyond those forms. What would we need to listen to, measure, or understand next? The projection cannot complete that work for us. It can help us choose to begin.

Original narration, not an artist quotation. References: [L1-L5], [S1-S11].

Presenter background / The factual foundation

What the project brings together

P.A.I.N. is an art, AI, and network-science collaboration. The exhibition's About panel names Mary Maggic as project lead and collaborators in art, design, sound, data research, software, AI, and the Ludwig Boltzmann Institute for Network Medicine. The acronym is Personal And Interconnected with Nature; PPP is Personal-Planetary-Pain. The supplied master PDF is a one-page recruitment brief that supports the broad artistic vision, not a methods paper. [L6]

The four layers are placed together as an interpretive artwork. Shared geography permits comparison, but neither co-location nor a shared 0-1 range establishes causation, equal measurement quality, or a scientifically valid sum. Environmental harm and economic hardship can matter to health and lived experience; the map alone does not estimate those relationships.

Complete coverage map for presenters

Topic	What to explain
Purpose and four layers	Artistic vision; separate indicators; combined viewing.
Sources	Publishers, counts, selection, countries, dates, languages, units.
Pipeline	Acquisition, cleaning, country coding, inference, summaries, visual encoding.
Emotional model	Synthetic training, real input, overlapping categories, controls, diagnostic status.
Language	Category meanings, chosen country languages, translations, winner simplification.
Physical	Six conditions, population-based placement, normalization and missing methods.
Environment	Warming and emissions are distinct; baselines, periods, and gas definitions matter.
Socioeconomic	GDP per capita, current US dollars, reversed logarithmic scale.
Visual grammar	Surface, dots, atmosphere, yellow hatching, labels, symbols, highlights.
Geography and summaries	Country/territory roster, simplified polygons, label points, maxima, no-data states.
Networks	Delaunay, Gabriel, category membership, great-circle drawing, animation.
Interpretation	Observation, construction, supporting context, and unresolved causal explanations.

Reading the four numerical layers

Layer	Quantity used by the exhibition	What it does not measure
Emotional	Weighted country means of model scores on selected texts.	Population prevalence or a national psychological diagnosis.
Physical	Normalized health-related spatial input; country symbol uses its maximum.	A directly recoverable patient count, rate, or national average.
Temperature	Normalized temperature-change input; country symbol uses its maximum.	Today's temperature or an independently verified national mean warming.
Emissions	Normalized emission-related input; country symbol uses its maximum.	Personal exposure, atmospheric concentration, or national total emissions.
Socioeconomic	Reversed logarithm of GDP per capita, 2024 current US dollars.	Wealth distribution, typical income, purchasing power, or directly measured suffering.

The physical and environmental source releases, input variables, and full transformations are not completely established. Calling an export normalized does not identify the formula that normalized it. For socioeconomic data, the formula is available and reproduced below. The emotional means are saved model scores, not a later within-country min-max heatmap. [L1-L4]

Source atlas / How much data, from where?

Four different counts

An upstream dataset may contain millions of records. PCAI admits a subset under its product rules. The classifier analyses selected rows. The display then reduces them to country scores or map points. None of those counts is automatically a number of unique people. The following tables label the stage explicitly. Unknown upstream totals are left unknown.

Emotional inputs actually analysed

Source	Analysed rows	Countries	Language hints	Recorded years
Stanford OVAL CCNews probe	5,000	186	59	2015-2021; 2023
Twitter Sentiment by Country	166,837	163	60	2020
DEMOTEC	67,193	136	9	2015-2021
Yachay Seasons	8,564	4	47	2014-2022
CFPB complaints	1,030	1	2	2025
CMU GeoText	814	1	8	2010
Five expression inputs pooled	244,438	175	68	Discontinuous 2010-2025
Six inputs combined	249,438	192	84	Discontinuous 2010-2025

Countries and language hints overlap, so their totals are unions rather than sums. Hints are metadata/detector labels, not a validation result. The project counts were recomputed from the selected metadata and inference products; they are not scraped source-page counters. The source dates are historical and discontinuous, not an exhibition-date survey. [L2]

S1 / Stanford OVAL CCNews

Published news text prepared from Common Crawl news archives. The provider describes roughly 600 million articles in more than 100 languages; that is an approximate upstream description, not this exhibition's sample size. It also distinguishes crawl year from publication date. The selected exhibition uses exactly 5,000 records from a bounded country probe. The newer 25,932,030-record classified news product is not selected here. News country means publisher organizational origin, not necessarily the location of a reported event. [S1, L2]

The probe includes 2,759 records with recorded year 2015 and 1,933 with 2016: $(2,759 + 1,933) / 5,000 = 93.84\%$. The remaining recorded years are 2017 (164), 2018 (75), 2019 (55), 2020 (11), 2021 (2), and 2023 (1). Date semantics and discrepancies with older acquisition descriptions remain a reason to call these recorded years, not verified publication dates. The sampling is a diagnostic design, not a representative national panel.

S2 / Twitter Sentiment by Country

The named source is the Kaggle dataset by wjia26. The admitted project input contains 166,837 rows, with supplied country context for 163 countries and recorded year 2020. Its 60 language hints describe the selected product. A complete upstream row total was not established for this guide. A supplied country field is not proof of author residence, nationality, or where the described event happened. This is the largest expression input by row count. [S2, L2]

S3 / DEMOTEC

The DEMOTEC research project collected tweets containing participatory-budgeting keywords in nine languages during 2015-2021. The public record identifies 24 variables and a codebook; its complete upstream row total was not verified here. The exhibition input contains 67,193 admitted rows across 136 country contexts. Geography comes from reviewed profile-location context. The civic topic and keyword collection matter: this is not a random sample of everything people said or felt. [S3, L2]

S4 / Yachay Seasons

The source card describes over 600,000 coordinate-linked social posts focused on six countries. The selected project input uses 8,564 rows across four country contexts, with 47 language hints and recorded years 2014-2022. Its country context is derived from coordinate bounding boxes. The broader source-card description and the admitted metadata are different stages; neither implies that each post has a precise verified author location. [S4, L2]

S5 / CFPB consumer complaints

The Consumer Financial Protection Bureau publishes a database of complaints about financial products and services. The selected input is 1,030 complaint narratives with one country context, two language hints, and recorded year 2025. The full database changes over time; this guide does not assert a fixed upstream total. The material is complaint-focused by construction. Complaint/event location must not be relabelled as a national mood measure. [S5, L2]

S6 / CMU GeoText

Carnegie Mellon's geotagged microblog corpus has a documented 2010 release. The selected project input contains 814 rows, one country context, eight language hints, and recorded year 2010. A full upstream row total is not established in this guide. Its geographic context is associated with users; this is not a claim about the precise location of each author or experience. [S6, L2]

Physical and environmental map inputs

Input	Exported map records	Observed polygon coverage	What remains unknown
Physical, all six conditions	15,282	155 of 177 geometries	Original health release, variable, strata, and full transformation.
Temperature	8,100	140 of 177 geometries	Exact downloaded product, reference implementation, period, and normalization.
Emissions, labelled CO2	2,339	146 of 177 geometries	Gas measure, period, sectors, original units, and normalization.

Coverage here uses all 177 available Natural Earth country/territory geometries and the application's point-in-polygon lookup. It differs from the 192-country emotional roster and the 206-entry profile union. Unassigned rows include ocean points and locations affected by simplified coastlines or small missing polygons, not only erroneous data. [L3, L5]

The six physical categories each contain 2,547 spatial rows. These are not 2,547 patients or 2,547 contributing studies. Source attribution names IHME GBD health estimates and WorldPop population data. The exact GBD release, measure (for example cases, rates, or disability), age/sex strata, and uncertainty have not been recovered. The WorldPop link identifies an approximately 1 km, UN-adjusted 2000-2020 product family, but not the actual year used or the resolution of the final health export. Country-to-grid placement and normalization remain documented project descriptions rather than independently reproduced upstream methods. [S7-S8]

Berkeley Earth provides gridded temperature and time-series products. The project's source panel describes a 2015-2025 warming average relative to 1850 and fixed scale references of 1 and 5 degrees Celsius. Those exact operations have not been verified against the original processing code. Do not invert the exported score into degrees, or assume a particular baseline or clipping rule, on the strength of that description alone. [S9, L3]

Climate TRACE publishes emission estimates with choices of gas, sector, and time interval. CO2 and CO2-equivalent are different quantities. The local category name does not establish which source measure was exported. The current map's 2,339 points must not be presented as Climate TRACE's complete inventory, a national emissions total, or measurements of local air quality. [S10, L3]

S11 / World Bank GDP per capita

The selected file records World Bank indicator NY.GDP.PCAP.CD for 2024, in current US dollars per person. It has 206 roster entries, 190 available values, and 16 null entries. Its metadata records an update date of 13 July 2026. That is distinct from the reference year 2024. The file replaces the older total-GDP endpoint for the selected exhibition display. [S11, L4]

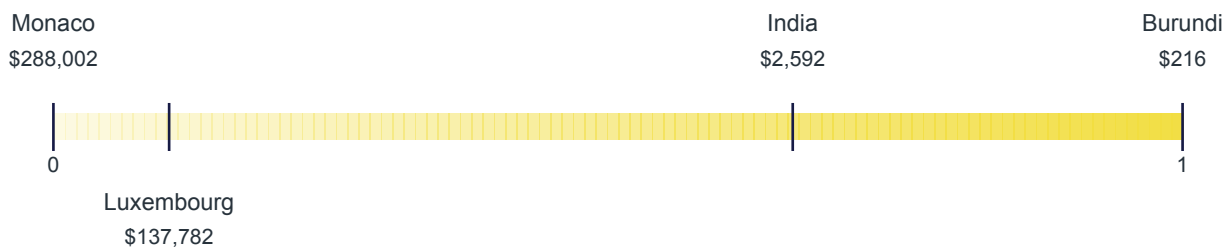
GDP describes production within an economy. GDP per capita divides it by population. It does not measure a typical salary or household income, and it does not adjust for purchasing power, inequality, or subjective well-being. Source metadata records CC BY 4.0 attribution terms; this guide links to the source rather than redistributing its full table.

How the economic scale works

For raw GDP per capita g , minimum a , and maximum b :

$$Y = 1 - (\ln(g) - \ln(a)) / (\ln(b) - \ln(a)).$$

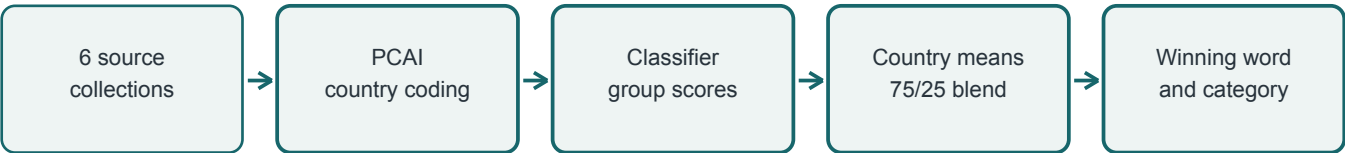
Here $a = 216.231928531758$ (Burundi) and $b = 288001.574856495$ (Monaco). The mapping gives Burundi $Y = 1$ and Monaco $Y = 0$. India's $g = 2591.9916607122$ gives $Y = 0.654754$, rounded to six decimals. A numerical endpoint of zero is not a claim of zero suffering. Missing GDP values do not enter this formula. [\[L4\]](#)



GDP per person decreases as the artistic yellow signal increases.

Methods / From text and points to the globe

The emotional pipeline



Training examples are separate from the real texts analysed here.

PCAI acquires, prepares, and country-codes admitted text. For news it uses publisher-domain reference information, a Public Suffix List, and usable country-code domain evidence. GDELT contributes publisher reference metadata; it is not a hidden seventh text collection in the selected emotional product. Expression records use their approved geographic context, with an admission confidence floor of 0.70. Missing country context is not simply filled from place names mentioned in prose. [L2]

pcai-classifier uses frozen BGE-M3 embeddings and a learned prediction head. The training-data program assembled 573,143 synthetic examples across 100 languages. That is the available package count, not the exact final-fit count for bundle-v9: its recipe excludes a legacy bridge source. Training examples must remain distinct from the 249,438 real-text inference inputs. The older small diagnostic model and the later full news run are different products.

The public group score composes a distress-scope score and a conditional group score: $P(\text{group}) = P(\text{distress scope}) \times P(\text{group given distress})$. Multiple groups can receive signal from the same text. These model outputs need not sum to one and are not clinical scores.

For country c and category k , let N be the mean over all selected news rows and E the mean over all selected expression rows pooled across their five source collections.

Available cohorts	Country score	Countries
News and expressions	$C = 0.75 N + 0.25 E$	169
News only	$C = N$	17
Expressions only	$C = E$	6
Neither, within the 195-country source roster	Missing	3

The denominator is all classified rows in the cohort, not only rows assigned a positive distress label. Public group scores are averaged even for rows whose decoded scope is uncertain, no-pain, or out-of-scope. The displayed table uses these raw weighted means, rounded to four decimal places. A separate analytical within-country min-max heatmap must not be substituted for it. All 3,072 observed values (192 x 16 columns) match the selected weighted calculation at the stored precision. [L2]

Fourteen categories, one visible word

Display label	Meaning covered by the classifier group	Winning countries
Body pain	General physical pain expressed in text.	3
Hurt	Emotional pain without a more specific description.	18
Eco-anxiety	Climate-related fear, anxiety, or panic.	1
Uncertainty	Instability and uncertainty.	1
Grief	Grief and solastalgia, including distress over a changed home environment.	3
Anger	Anger and moral injury, including outrage at perceived wrongs.	152
Hardship	Loss and precarity.	6
Displacement	Exile and displacement.	1
Trauma	Trauma and overwhelm.	1
Loneliness	Relationship and social pain.	0
Depression	A broad group including depressive distress and suicidal-ideation language.	0
Fear	General fear, anxiety, or panic.	3
Helplessness	Helplessness and powerlessness.	0
Shame	Shame, guilt, and self-blame.	3

Counts describe the selected data with all categories enabled. Zero winners does not mean zero underlying score or no such experience in the world. The grouping is a classifier taxonomy; it is not a clinical diagnosis and its brief display words compress broader meanings. No visitor is being assessed by these country labels.

[\[L1-L2\]](#)

The display excludes no_pain and out_of_scope when selecting its strongest word. It also reranks among remaining categories when a visitor excludes one. The underlying scores are unchanged by that filtering. A larger word can reflect camera distance; brightness can reflect selection and visibility. Neither alone ranks emotional severity. [\[L5\]](#)

The lexicon chooses a country language and script. Native terms and supporting English translations help visitors read across languages, but a country is often multilingual and a short English gloss is not a complete cultural interpretation. The selected winning terms have no English fallback entries; this does not establish equal translation quality.

What the model validation does and does not establish

The saved selected predictions have `accepted_model=false` and `real_world_validated=false`. They are diagnostic outputs. The training roster includes 100 languages, while the saved release declaration lists English, Spanish, and Arabic. Wider processing is best-effort diagnostic use, not verified equal accuracy across the language roster. Of the news probe, 2,605 rows have language hints in that declared set and 2,479 were truncated at the documented 512-token input limit. A token is a model's text unit, not always a complete word. [L2]

These facts do not supply a measured error rate, a national gold standard, or confidence intervals for country ranks. A large sample cannot eliminate model bias, unrepresentative sources, or incorrect geography. The current score comparisons should be narrated as properties of this selected corpus and model.

Spatial profiles and visual fields

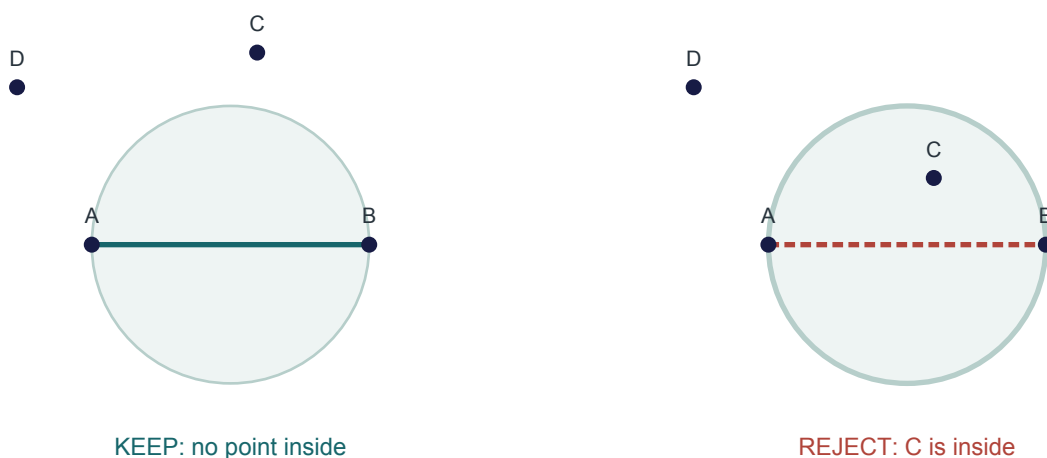
For physical, temperature, and emissions data, the app assigns coordinate points to its Natural Earth polygons and selects the maximum relevant value. It retains the contributing point count. Physical pools all six condition categories for that maximum. Temperature and emissions remain separate. A point count describes export coverage, not patient sampling. Simplified boundaries can affect assignment near coastlines and small countries. [L3, L5]

The globe surface and the country symbols are separate renderings. Physical points stamp overlapping depressions into a field that is smoothed. Temperature and emissions use their own accumulation, smoothing, and appearance controls. A rendered haze cannot be read as a linear ruler of the original source variable. Country area symbols use a square-root size mapping so their filled area represents their normalized signal; the emissions band has its own treatment. [L5]

Country locations for words and graph nodes use cartographic label points and some manually specified small-country positions. They are not necessarily geometric centroids. The geography/profile union has 206 entries, whereas the emotional input has 192 observed countries. Keep those denominators separate. A missing symbol may reflect unavailable data or geography; do not describe every blank location as an observed zero.

The network, precisely enough to explain

Gabriel candidate-edge test



2D schematic; the actual implementation tests a 3D diameter ball.

Each observed country's selected emotion assigns it to a category group. Its geographic label position becomes a unit vector on the globe. For four or more nodes, the current Delaunay rule builds a three-dimensional convex

hull and uses the unique edges of its faces. The renderer draws curved great-circle arcs above the surface. [L5]

Gabriel mode starts from those candidate edges. For endpoints a and b , take midpoint $m = (a + b) / 2$ and squared radius $r2 = \text{distance}(a,m)^2$. An edge is rejected if another point c in the same category satisfies $\text{distance}(c,m)^2 < r2 - 1e-12$. It is an empty three-dimensional diameter-ball test, not a test on screen-space text positions. Points on the boundary do not reject the edge. [L5]

Implementation qualification: for fewer than four nodes, the app returns all pairs even in Gabriel mode.

Near-coincident or near-antipodal endpoints can also be omitted by the arc renderer. Do not quote historical graph-edge counts as current: category membership and filters change the graph. The current selected graph is Delaunay, not Gabriel.

The reveal sequence uses graph steps outward from the selected origin. This is interaction animation, not a temporal model. Lines do not encode correlation, influence, transport, friendship, shared exposure, or transmission. Geographic proximity and category membership are enough to construct them; no interpersonal or causal evidence is required.

Practical exhibition operation

Begin with all layers and orient the globe before introducing numbers. Select the emotional layer to expose its category controls, then a category to reveal its network. Repeat selection can clear the network. Excluding a category reranks the remaining categories, so explain the change in question before discussing the new pattern. Restore all categories before quoting the winner counts in this guide. [L5]

The projection mode hides the visitor-survey action. The related web mode can offer a survey and generated response, which is a separate participation flow and not a seventh source in the displayed country dataset. Do not ask a projection audience to disclose personal health information or describe an absent interface control as available.

Unresolved methods to keep visible

- Physical: original GBD release, year, variable, age/sex strata, uncertainty, population raster year, allocation, and normalization.
- Temperature: original Berkeley Earth file, baseline calculation, exact temporal selection, scaling, and treatment of values outside the described reference range.
- Emissions: original Climate TRACE release, gas, units, time period, sectors, and scaling.
- Emotional: complete publication/crawl-date semantics, source representativeness, and validation of the intended country-level interpretation.

These are targeted information gaps. They do not prevent explaining the inspected application, but they limit physical-unit conversions and causal country claims.

Story collection / 25 ways to begin a conversation

Choose a few stories for a visit, not all 25. The Say paragraphs are original presenter wording. Evidence and Keep in mind are background notes. These stories extend the nine complete scripts. All map values refer to the selected exhibition data checked on 9 September 2026. Spatial comparisons use the same 177 country/territory geometries as the application. Near-minimum examples exclude missing observations; disputed territories are not presented as sovereign-country rank winners. Numbers compare the named signal only. [L1-L5]

Story 01 / India: one place, four different questions

Say: India can look intense on this globe, but there is no single number called India's pain. Its physical symbol takes a local maximum from a migraine-labelled point. Its emotional word comes from selected texts. Its warming and emissions symbols use other local maxima. Its yellow signal comes from economic output per person. Seeing them together invites a question about relationships. It does not prove that one caused another. The first useful explanation for the physical concentration is how country-based health information was placed using population data. Understanding medical causes would require the original health measure too.

Evidence: Physical peak 0.94266, 266 spatial records, with the peak in Migraine near 19.105 N, 72.913 E; temperature peak 0.151085 from 31 points; emissions peak 0.25547 from 40 points; GDP per capita US\$2,591.99; emotional anger score 0.2331 from 27 news and 1,442 expression records. [L2-L4]

Keep in mind: A mapped concentration is not an individual risk estimate or a national mean.

Story 02 / Pakistan: two headache labels, not two patient totals

Say: Pakistan has a high mapped peak in both Headache disorders and Migraine. That gives us a chance to ask what a category means before adding numbers. Migraine and a broader headache category should not automatically be treated as separate groups of people. Here the country profile takes a maximum; it does not add the two condition surfaces. A value of one means reaching an endpoint of the exported normalized surface. It does not mean that everybody has the condition, or that we know Pakistan's national rate from this image.

Evidence: Both condition peaks are 1.00000, each based on 13 assigned condition points. The combined physical profile receives 82 records. [L3]

Keep in mind: The exact upstream health hierarchy and normalization are unresolved.

Story 03 / Japan: back pain and the question of age

Say: Japan's low-back-pain surface reaches a normalized peak of one. A tempting explanation would immediately connect that result to age. WHO does identify age and population growth as important to the global burden of low back pain. But that context does not establish why this particular Japanese point is high. We first need the original dataset's age structure, whether it reports counts or rates, and how population was used in mapping. The story teaches us to turn a plausible explanation into a question that can be checked.

Evidence: Low back pain peak 1.00000 from 17 condition points; 102 records contribute to Japan's all-condition physical maximum. WHO context is general, not an analysis of this map. [L3, C3]

Keep in mind: Do not present ageing as a demonstrated cause of Japan's displayed rank.

Story 04 / China: arthritis is not one condition

Say: China's high physical values include both osteoarthritis and rheumatoid arthritis. The shared word arthritis can hide differences. Osteoarthritis affects the joint and is associated with factors including age and injury. Rheumatoid arthritis involves immune-driven inflammation. Their routes into a health dataset are not interchangeable. The globe's country symbol compresses the condition surfaces into one maximum. Opening the story back up means asking which condition produced a value and what evidence would explain that condition.

Evidence: Both mapped condition peaks reach 1.00000; osteoarthritis has 135 assigned points and rheumatoid arthritis 137. The combined physical profile has 803 records. [L3, C5-C6]

Keep in mind: These are spatial export counts, not patient totals or disease prevalence.

Story 05 / Egypt: a strong neck-pain point

Say: Egypt is a useful example of how a very high-looking score can tempt us into a story we have not verified. Its neck-pain surface reaches almost one. That observation alone does not identify posture, work, treatment access, or any other cause. The country's shape on the globe is a summary of placed and normalized health information. The honest next step is to ask for the source measure and its mapping method. The visual pattern gives us a reason to investigate, not the answer to the investigation.

Evidence: Neck pain peak 0.99900 from 12 assigned neck-pain points; 72 records contribute to the overall physical profile. [L3]

Keep in mind: Possible occupational or lifestyle explanations have not been tested here.

Story 06 / United Kingdom: precision is not certainty

Say: The United Kingdom has a rheumatoid-arthritis point with a normalized value very close to one. We could print five decimal places, but that would not give us five decimal places of medical certainty. Precision in an exported file describes how a number is stored. Confidence in a health conclusion depends on the source data, model, and uncertainty. This story is useful whenever a visitor asks whether two nearly equal map values really mean that one country's residents suffer more than another's.

Evidence: Rheumatoid arthritis peak 0.99961 from nine condition points; 54 records in the combined physical profile. [L3]

Keep in mind: The export supplies no verified uncertainty interval for this comparison.

Story 07 / Guyana and Vanuatu: what a low peak leaves out

Say: Guyana and Vanuatu have low observed physical peaks in this export. We should resist turning that into a statement that people there have little pain. Each profile is based on a small set of placed values and a maximum rule. Source coverage, normalization, and the choice of conditions all influence what can become visible. Low signal is an observation about this layer. A fuller account of health would require rates, age patterns, local evidence, and conditions that may not be represented by these six surfaces.

Evidence: Guyana's combined physical peak is 0.00073 and Vanuatu's 0.00202, each from 12 spatial records. Both reported maxima are Migraine points. [L3]

Keep in mind: These low values are observed values, not missing observations.

Story 08 / Russia: a cold place can warm strongly

Say: Russia's high temperature-change profile helps separate two questions: how cold is a place, and how much has it changed? A place can still be cold while warming substantially. IPCC assessments describe amplified Arctic warming through several interacting processes, including changes in snow and ice reflectivity. That is relevant regional context. This country profile, however, selects the largest mapped value inside Russia. It is not a national average, and its normalized scale cannot yet be converted confidently into degrees from the original processing evidence available to us.

Evidence: Temperature peak 0.7058875 from 363 assigned points, the highest country/territory profile peak in the current comparison. [L3, C1]

Keep in mind: Large country area and many observed locations matter to a maximum comparison.

Story 09 / Canada: many locations, one selected maximum

Say: Canada's profile also has a strong warming signal. The compact symbol gives us one value, while the calculation considers many locations. Selecting the maximum emphasizes the strongest observed place, rather than an average resident's surroundings or a national mean climate. This is a useful choice for drawing attention to extremes, but it changes the comparison. A country with many contributing points has more opportunities to contain an extreme. We need the underlying distribution to know how typical the maximum is.

Evidence: Temperature peak 0.614436875 from 220 assigned points, second in the current country/territory peak comparison. Arctic warming provides context, not a pointwise attribution. [L3, C1]

Keep in mind: The globe's visible atmospheric field is another transformation of the points.

Story 10 / Uruguay: zero on this scale

Say: Uruguay's observed temperature profile is zero in this normalized export. That is not a finding that climate change has had no effect there. Zero may be an endpoint created by a transformation, and we have not recovered all the steps that produced this temperature scale. We can say exactly what the application reads, while leaving the broader climate question open. This is a helpful example of why a numerical zero and a real-world absence are not always the same statement.

Evidence: Maximum of two assigned Temperature records is 0.00000. [L3]

Keep in mind: Do not assume a particular clipping threshold without the processing code.

Story 11 / Iceland: a tiny value is not a climate verdict

Say: Iceland's temperature peak is very small in this export. A visitor familiar with regional climate change might find that surprising. Surprise is useful here: it tells us to inspect the product, locations, baseline, and normalization before writing a national story. The profile summarizes three assigned points. It does not replace a national climate time series. A small symbol can invite us to look more carefully at what the data pipeline has retained, rather than conclude that a familiar scientific picture must be wrong.

Evidence: Temperature peak 0.00006609375 from three points. [L3]

Keep in mind: This guide does not identify the cause of the unexpectedly small export value.

Story 12 / Russia's emissions: a hotspot is not a country total

Say: Russia reaches the high endpoint of the local CO₂-labelled export. The country symbol keeps a spatial maximum. It does not add all emissions within the country, divide them by population, or reconstruct a historical total. Those would be three different comparisons. Climate TRACE offers data across gases, sectors, and periods, so the missing export details are important. The dramatic atmosphere can prompt a question about an emissions hotspot; it does not settle which country emits most overall.

Evidence: CO₂-labelled peak 1.00000 from 273 assigned points. [L3, S10]

Keep in mind: Do not identify a facility or sector from an approximate plotted point alone.

Story 13 / Iran and Iraq: different maxima, different coverage

Say: Iran and Iraq both have relatively high emissions peaks in this exported layer. Their point counts differ, and the country symbol chooses only the largest value. We cannot turn those maxima into national totals or explanations about a particular industry without the source release, gas definition, period, and sector information. The useful story is how two nearby countries can invite a comparison while still requiring us to define precisely what is being compared. Geographic closeness does not repair missing measurement details.

Evidence: Iran's CO₂-labelled peak is 0.48880 from 22 points; Iraq's is 0.41472 from five. [L3]

Keep in mind: The difference is in a normalized local-maximum statistic, not tonnes per person.

Story 14 / Guinea-Bissau: one low emissions observation

Say: Guinea-Bissau is near the lower end of the observed emissions profiles. Its value is supported by only one assigned export point. That does not establish an inventory of every activity in the country or prove that the surrounding air is clean. One observation can be useful, but it cannot answer all the questions that a whole-country symbol suggests. This is a good place to distinguish emissions estimates from exposure and to ask how spatial coverage changes the meaning of a map.

Evidence: CO₂-labelled peak 0.17041 from one point. There are no zero-valued points in the 2,339-record emissions export. [L3]

Keep in mind: The nonzero lower range is not independently explained by the available transformation evidence.

Story 15 / Burundi: why economic context matters

Say: Burundi sets the strong-yellow endpoint because it has the lowest GDP per person among the available values in this selected file. The World Bank describes a landlocked economy with substantial low-productivity agricultural activity and constraints on economic development. That gives context for the production statistic. It does not tell us how every household lives, how resources are distributed, or what one person feels. We can explain the economic conditions while avoiding a claim that yellow measures individual suffering.

Evidence: GDP per capita US\$216.23 in 2024; inverse-log yellow signal 1.000000. [L4, C2]

Keep in mind: The economic source explains output and constraints, not a validated pain index.

Story 16 / Luxembourg: production is not a typical salary

Say: Luxembourg's high GDP per person produces a relatively weak yellow signal. Its national statistics office explains why GDP per resident needs special care: cross-border workers contribute to the economy's production while living outside the resident population used in the denominator. The statistic remains meaningful as defined, but it is not a typical resident's salary. This is a concrete example of how a country's economic structure can affect a comparison before we say anything about quality of life or suffering.

Evidence: GDP per capita US\$137,781.68; yellow signal 0.102482. The explanatory accounting principle comes from STATEC; its older publication is not the source of the 2024 value. [L4, C4]

Keep in mind: Household income and distribution would answer different questions.

Story 17 / Monaco: the endpoint that becomes zero

Say: Monaco has the largest GDP-per-person value in this selected file, so the reversed scale gives it a yellow signal of zero. That result comes directly from the formula's endpoint. It does not mean an absence of suffering. In fact, the independent emotional dataset assigns Monaco a visible hurt label. Putting those facts together does not create a contradiction: the two layers use different sources, definitions, and scales. This is why we should never describe a zero in one layer as a complete judgement of a place.

Evidence: GDP US\$288,001.57; Y = 0. Emotional hurt score 0.1309 from 27 news and 1,034 expression rows. Tiny-country geometry can limit surface display. [L2, L4-L5]

Keep in mind: The model's hurt label is not a population survey.

Story 18 / India: GDP and the denominator

Say: A country's total economy and its economic output per person are different views. The exhibition uses the per-person measure. India's value is about 2,592 current US dollars for 2024, producing a yellow signal around 0.655 after the reversed logarithm. That does not mean 65.5 percent of people are poor, nor that a typical person earns that amount. The story is about choosing a denominator before drawing a colour. Changing the source to total GDP would change the question and the image.

Evidence: GDP per capita US\$2,591.99; Y = 0.654754; 46th highest yellow signal among 190 available values. [L4]

Keep in mind: Current US dollars do not adjust the figure for local purchasing power.

Story 19 / Yemen: missing in one layer

Say: Yemen has no available GDP value in this particular 2024 file. The correct response is to show it as unavailable, not to fill it with zero and call the country economically comfortable. Another layer can still contain observations: the emotional product has a grief label for Yemen. These are different source products with different coverage. A blank area tells us to ask about missing information. It should not be interpreted as a statement that a country is unaffected by hardship.

Evidence: GDP entry is null. Emotional winning grief score is 0.1886 in the selected JSON. [L1, L4]

Keep in mind: The source's missingness reason is not established by the null itself.

Story 20 / Burundi's anger score: a high value, a small sample

Say: Burundi has the highest anger score in the selected emotional table. Before offering a political or cultural explanation, look at what actually produced it: 28 news records and two expression records. The news mean receives three quarters of the weight, and the two-expression mean receives one quarter. This is a high result for a particular corpus and model. It cannot justify describing Burundi's people as the angriest. The important story is how a striking numerical extreme can rest on very little selected material.

Evidence: Anger 0.5202; 28 news rows and two DEMOTEC rows. [L2]

Keep in mind: Contextual national events would not by themselves prove the model's interpretation.

Story 21 / Palau: the danger of an easy low-score story

Say: Palau has the lowest anger score in this table, but it would be misleading to call it the calmest country. The result comes from one selected news record and no expression records. Its displayed word is actually uncertainty, because that is its largest retained distress category. An out-of-scope score is higher still, but the display excludes that control from choosing the word. This example reveals the difference between a record, a model output, a selection rule, and a story we might too quickly tell about people.

Evidence: Anger 0.0105; uncertainty 0.1557; out_of_scope 0.6492; one news row. [L2]

Keep in mind: A minimum model score is not evidence of national emotional well-being.

Story 22 / Timor-Leste: the influence of a few texts

Say: Timor-Leste is also near the high end of the anger scores. Its selected evidence is five news records and three expressions. The weighted calculation can be reproduced, but that does not make the sample representative. A small number of strongly scored texts can have a large effect on a mean. Before looking for a national cause, we should ask which sources, topics, dates, and language conditions those few records reflect. The map gives us a place to start that inquiry, not a shortcut around it.

Evidence: Anger score 0.4738; five news rows and three DEMOTEC rows. [L2]

Keep in mind: Do not confuse this with Timor-Leste's separate zero temperature profile.

Story 23 / Sao Tome and Principe: climate language is not a climate sensor

Say: Sao Tome and Principe has the highest eco-anxiety score in the selected text table. That can lead to an interesting conversation about how environmental concerns appear in language. It cannot establish that residents there experience the highest climate anxiety, or that the country has the most severe physical climate hazard. This score comes from 27 publisher-linked news records and no expressions. The emotional and environmental layers can be considered together, but they are measuring and constructing different things.

Evidence: Climate-fear/anxiety/panic score 0.2008; 27 news records, zero expressions. It is the sole eco-anxiety winner with all categories enabled. [L2]

Keep in mind: News publisher geography is not necessarily the location described in an article.

Story 24 / Oman: the same word, a different source

Say: Oman's visible emotional label is body pain. It is tempting to assume it came from the separate health layer, but it did not. It came from a classifier scoring text. The physical surface comes from health estimates and population-based placement. Two layers can use related words while describing different evidence. That makes Oman a useful example whenever someone asks why a word and a physical mark do not seem to agree. We first need to compare like with like before treating their difference as a finding.

Evidence: General physical pain in text score 0.1580; 27 news and 968 Kaggle expression rows. This is the highest score for that text category in the selected table. [\[L2\]](#)

Keep in mind: It is not a clinical prevalence estimate or the six-condition physical maximum.

Story 25 / Comoros: absence is specific to the dataset

Say: Comoros has no observation in the selected emotional product, but it does have an economic value. This is a simple way to explain why a single blank on the globe cannot mean that a place is invisible to every source or has no suffering. Different datasets reach different places in different ways. The useful question is which layer is missing information and why. A careful exhibition explanation keeps that gap open instead of filling it with an invented national story.

Evidence: Zero news and expression rows in the selected product; emotional entry missing. GDP per capita US\$1,857.87; yellow signal 0.701039. [\[L2, L4\]](#)

Keep in mind: Small-country polygon availability is a further, separate display limitation.

Questions visitors may ask

Does the globe show which country suffers most?

No single ranking is established. A meaningful comparison names a layer, its source variable, time period, sample or point count, and aggregation. The layers do not share a validated unit of suffering. Some source transformations remain unverified.

Why are so many words anger?

With all categories enabled, anger is the highest retained score in 152 of 192 observed countries. That is a property of this selected corpus, weighting, model, and winner rule. It does not mean most residents are angry. Source composition and classifier behavior may contribute, but their causal contributions have not been measured here.

Are the lines Gabriel graphs?

The selected exhibition uses Delaunay graphs. Gabriel is a related option implemented during development. Its empty-diameter-ball test removes some Delaunay candidate edges in the usual four-or-more-node case. Both are geographic constructions inside category groups.

Why does a word change when a category is excluded?

The display chooses the highest remaining category. Scores have not changed; the set of eligible answers has. Category membership and therefore the graph can change too.

Is this live data?

It is a saved selected product. The text sources span uneven historical years, GDP refers to 2024, and precise health/environmental source releases remain partly unresolved. The date of opening the website or exporting this guide is not the observation date.

Does GDP per person mean personal wealth?

It is economic production divided by population. It does not measure wealth holdings, a typical wage, income distribution, or purchasing-power-adjusted living standards.

Can I infer a disease or emotional condition about somebody from this?

The country layers do not assess individual visitors or identify patients. An individual's health and experience require a different kind of evidence and conversation.

Glossary / Words worth keeping close

Term	Plain explanation
Aggregation	Combining many observations into a summary, such as a mean or maximum.
Baseline	A reference condition used to calculate a change.
Classifier	A model that assigns category scores to an input.
Corpus	A collection of texts selected for analysis.
Convex hull	The smallest convex shape enclosing a set of points, like a tight shell around them.
Denominator	The quantity being divided by; it changes what a rate or average means.
Diagnostic output	An exploratory result whose intended real-world interpretation is not validated.
Delaunay	A proximity construction; this app uses edges of the convex hull of spherical points.
Gabriel	A proximity rule using an empty ball whose diameter joins two candidate endpoints.
Inference	Applying a trained model to new input.
Language hint	Metadata about likely language, not proof of model accuracy.
Logarithm	A transformation that makes multiplicative differences easier to compare.
Maximum	The largest observed value, not the average.
Missing	An observation is unavailable; it is not automatically zero.
Normalization	A numerical rescaling; the formula and reference range must be specified.
Per capita	Per person, using the relevant population denominator.
Prevalence	The proportion or number of people with a condition in a defined population and period.
Proxy	A variable used to stand in for a broader concept it does not measure completely.
Sampling bias	A selected collection differs systematically from the population of interest.
Solastalgia	Distress associated with unwanted change to a familiar home environment.
Spatial allocation	Placing aggregate information across geographic locations using a rule or model.
Synthetic training data	Constructed examples used for learning, distinct from observed input records.
Token	A model's unit of text, which may be a word part or punctuation.
Uncertainty	What remains unresolved about a value or interpretation, beyond its printed precision.

References / Public sources and project evidence

References were checked on 9 September 2026. Provider descriptions explain their datasets; project evidence establishes which selected inputs and transformations the exhibition uses. A provider's present-day page does not prove the release used by an older project export. Country-story context is cited separately from the displayed country values.

Public dataset sources

- [S1] Stanford OVAL CCNews.** Dataset card, upstream scope and crawl/publication-date distinction. [Source page](#).
- [S2] Twitter Sentiment by Country, wjia26.** Source identifier from the acquisition records. The source page did not expose a usable full count during verification. [Source page](#).
- [S3] DEMOTEC participatory-budgeting tweets.** Zenodo record, version 1, 4 February 2025; collection topic, languages, and 2015-2021 period. [Source record](#).
- [S4] Yachay Seasons.** Coordinate-linked social-media dataset card and upstream scope. [Source page](#).
- [S5] CFPB Consumer Complaint Database.** Institutional explanation of the complaint collection. [Source page](#).
- [S6] CMU GeoText.** Geotagged microblog corpus; release 2010-10-12 and associated research. [Source page](#).
- [S7] IHME Global Burden of Disease.** Health-estimation study. The exact project release is unresolved. [Source overview](#).
- [S8] WorldPop Global 2000-2020.** Population product family; UN-adjusted approximately 1 km data. [Product overview](#). [Project-attributed density directory](#).
- [S9] Berkeley Earth.** Temperature datasets and product descriptions. [Data overview](#).
- [S10] Climate TRACE.** Emissions data and measurement definitions. [Data downloads](#). [Inventory definitions](#).
- [S11] World Bank.** GDP per capita, current US dollars, indicator NY.GDP.PCAP.CD. [Indicator](#). [Selected-year API query](#).

Context for selected stories

- [C1] IPCC AR6 Working Group I, Chapter 4.** Regional amplification and the interacting processes underlying Arctic warming. General regional context, not a reconstruction of this export. [Chapter](#).
- [C2] World Bank, Burundi overview.** Economic structure, landlocked geography, and development constraints. Accessed 9 September 2026; contextual overview, not a decomposition of the 2024 GDP figure. [Country overview](#).
- [C3] WHO, Low back pain, 19 June 2023.** Age patterns and global burden; not a Japan-specific explanation of the exhibition's mapped maximum. [Fact sheet](#).
- [C4] STATEC, Luxembourg in figures and analysis, Kaleidoscope 2006.** Explains the GDP-per-capita effect of cross-border workers. Used for the accounting principle, not current numerical estimates. [Publication](#).
- [C5] WHO, Osteoarthritis, 14 July 2023.** Condition definition and general risk factors. [Fact sheet](#).
- [C6] MedlinePlus, Rheumatoid arthritis.** Immune-related condition definition. [Medical encyclopedia](#).

Local exhibition evidence

Local paths below are relative to the exhibition frontend repository unless stated otherwise. They are maintenance references for the project team. Public visitors do not need a local checkout to understand the scripts. Source prose from individual posts or complaints is not reproduced in this guide. All stories are original explanations of aggregate observations.

[L1] Selected display data. `src/countryProfile/presets.ts` selects `v48-a_web`, `combined-v2`, and `gdp-per-capita-2024`. `data-src/country-emotion-combined-v2.csv` and `public/emo/combined-v2/emo-data.json` supply the selected emotional values. The JSON was generated 8 September 2026 from the selected 5 September v2 product. Values matched the live local exhibition during verification.

[L2] Emotional acquisition and model evidence. The selected country table is `products/country-emotion-analysis/v2/20260905T063105Z/tables/combined-ccnews75-expression25-mean-all-labels.csv` under the external PCAI data root. News predictions are under `tmp/country-classifier-real-probe/run-20260904T182154Z/`; expression metadata is in `products/country-classifier-pilot/v2-target-10000/20260822T213703Z/records.parquet`. The paired expression summary is `tmp/country-classifier-expression-runs/expression-summary-bundle-v9.json`. In `pcai-classifier`, `scripts/build_country_emotion_analysis.py`, `src/pcai_classifier/classifier/v2_decoder.py`, and the multilingual-training progress record explain aggregation and model maturity. In the active PCAI data worktree, `docs/reference/country-classifier-datasets.md` and `src/pcai/config/country_classifier_export_v1.yaml` define country attribution and input policy.

[L3] Spatial input and profile calculation. Read-only local `/init/phypain` and `/init/envpain` arrays checked 9 September 2026. `src/countryProfile/data.ts` and `src/globe/countryGeometry.ts` define maximum aggregation and point assignment using `public/borders/ne_110m_admin_0_countries.geojson`. All 1,338 country/condition values and contributing-point counts used in the comparison inventory were recomputed against fresh arrays. `src/ui/infoModal.ts` supplies the project-attributed upstream source descriptions.

[L4] Socioeconomic values and formula. `public/emo/gdp-per-capita-2024.json`, `scripts/build-gdp-per-capita.mjs`, and `src/countryProfile/gdpPerCapita.ts`. The selected file has 190 values and 16 nulls across 206 roster entries. It is separate from the 204-row legacy total-GDP API payload.

[L5] Visual and network behavior. `src/emo/graphs.ts`, `src/emo/arcs.ts`, `src/emo/categoryFilter.ts`, `src/emo/labelLayer.ts`, `src/emo/spread.ts`, `src/api/countryCentroids.ts`, `src/emo/viewPresets.ts`, and the country-profile and globe rendering modules. Current screenshots were captured from projection mode on 9 September 2026. The source checkout is `feat/exhibition-web`, based on commit `6c9b3b0`, with concurrent visual edits present. This guide does not claim those unrelated edits as its own implementation.

[L6] Project vision. Exhibition `src/ui/infoModal.ts`, and the supplied PAIN Project Master Doc.pdf recruitment brief. The brief introduces the artistic collaboration and intended projection context; it contains no detailed data methodology.

Keeping this edition useful

Before reusing a numerical story after a dataset change, check its metric, source, denominator, and date against the new selection. A later classifier run or a changed category filter can produce different labels. Preserve the distinction between a change in evidence and a change in visual treatment. Additional source-method material can improve the explicitly open sections without rewriting supported observations into stronger claims.